

Violence Against Civilians and Leader Favorability in Mali

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2026-10-06

1 Question and data

Is violence against civilians in a respondent's commune associated with how favorably they view the national leader, comparing people in the same region, urban or rural area, and survey wave? Violence is the number of ACLED events in the commune during each wave's survey window. About eight in ten respondents view the leader favorably, so any association is expected to be modest. The model sequence and decision rules were fixed before the violence coefficient was examined.

The analysis file has 3,264 respondents in 180 communes and 48 cercles. Each wave is a separate sample: communes are drawn within 15 strata (region by urban or rural), with eight respondents per commune.

Table 1: Sample by wave: design-weighted percent favorable (design-based SE) and median events per respondent's commune.

Wave	Respondents	Communes	% favorable	SE	Median events
1	1088	136	77.8	1.9	1
2	1088	136	81.3	1.4	1
3	1088	136	75.2	1.9	1

2 Models

2.1 Measures

For commune c in wave w with E_{cw} events, violence is $V_{cw} = \log(1 + E_{cw})$. Zero events gives 0; adding 1 multiplies $(1 + E)$ by $e \approx 2.72$, and adding $\log 2 \approx 0.69$ doubles it. Results are also given as predicted favorability at 0, 1, 5, 20 events. Respondent covariates $\mathbf{x}_i$: female, age, ethnicity, education.

2.2 Model sequence

Model	Adds	Compared by
M1a	commune random intercept	
M1b	adds a cercle random intercept	M1b vs M1a: likelihood ratio test with boundary correction, AIC, BIC
M2a	violence, wave, region, urban, covariates	
M2b	the strata in place of region and urban	M2b vs M2a: likelihood ratio test, AIC, BIC
M3	the chosen structure plus the centered log survey weight	final model; design-based check with svyglm
Sensitivity	violence split into cercle average and within-cercle difference	

All models are logistic multilevel models (`lme4::glmer`, maximum likelihood with the Laplace approximation), unweighted, on the same respondents, so likelihoods and AIC/BIC are comparable. The cercle level is kept if its test gives $p < .05$, AIC is lower, and the fit is not singular; the strata replace region and urban if $p < .05$ and AIC is lower.

2.3 Equations

Respondent i in commune c , cercle k , wave w ; $y_{ickw} = 1$ for a favorable view.

M1a. $\text{logit Pr}(y = 1) = \gamma_0 + u_c$, with $u_c \sim N(0, \tau_c)$.

M1b. $\text{logit Pr}(y = 1) = \gamma_0 + u_c + r_k$, with $r_k \sim N(0, \tau_k)$: communes nested in cercles.

A logit model has no residual variance; on the latent scale the individual variance is $\pi^2/3 \approx 3.29$, so

$$\rho_{\text{cercle}} = \frac{\tau_k}{\tau_k + \tau_c + \pi^2/3}, \quad \rho_{\text{commune}} = \frac{\tau_k + \tau_c}{\tau_k + \tau_c + \pi^2/3}$$

ρ_{cercle} is the correlation between two people in the same cercle but different communes; ρ_{commune} between two people in the same commune. The cercle test is $H_0 : \tau_k = 0$; because a variance cannot be negative, the p-value is half of the χ^2_1 p-value from `anova()`.

M2a.

$$\text{logit Pr}(y = 1) = \gamma_0 + \beta V_{cw} + \sum_{w=2}^3 \delta_w \text{Wave}_w + \sum_{r \neq \text{ref}} \alpha_r \text{Region}_r + \lambda \text{Urban} + \mathbf{x}_i' \boldsymbol{\theta} + u_c + r_k$$

The reference region is Bamako and the reference wave is wave 1.

M2b. The same, with $\sum_r \alpha_r \text{Region}_r + \lambda \text{Urban}$ replaced by $\sum_{s \neq \text{ref}} \kappa_s \text{Stratum}_s$. M2a is nested in M2b, so the test asks whether the urban-rural gap differs by region.

M3 (final). The chosen structure plus $\phi \log \tilde{w}_i$, the centered log of the survey weight. Strata are fixed effects and communes the random intercept; `glmer` cannot use sampling weights, so the weight enters as

a covariate. A clearly nonzero ϕ would mean people with larger weights differ in favorability beyond the predictors (informative sampling; Pfeffermann et al., 1998; Gelman, 2007).

Design-based check. The same fixed effects fit with `survey::svyglm`: weights rescaled within wave, strata within wave, communes as primary sampling units, Taylor-linearized standard errors (Binder, 1983). Its coefficients are population-averaged and smaller than the multilevel ones by roughly $\sqrt{1 + 0.346 \tau}$ (Zeger, Liang & Albert, 1988), so it is compared by z .

Sensitivity (cercle split). V_{cw} is replaced by V_{kw}^- , the average across the surveyed communes of cercle k in wave w , and $V_{cw} - V_{kw}^-$ (Mundlak, 1978). The within-cercle coefficient compares communes in the same cercle and wave, so stable cercle context cannot explain it.

3 Results

3.1 How much favorability clusters

Table 2: Latent-scale intraclass correlations from the empty models.

Model	Level	ICC
M1a	Commune	0.159
M1b	Cercle (same cercle, different commune)	0.057
M1b	Commune (same commune, includes cercle)	0.166

In M1b, 16.6% of the variation in the latent propensity to view the leader favorably lies between communes, of which 5.7 percentage points are shared across communes of the same cercle.

3.2 Model comparison

Table 3: Fit of the model sequence (same respondents, maximum likelihood). Tests compare M1b with M1a and M2b with M2a.

Model	Log-lik.	Parameters	AIC	BIC	Test	Singular
M1a	-1681.82	2	3367.64	3379.83		FALSE
M1b	-1678.70	3	3363.40	3381.67	chi2(1) = 6.25, p = 0.006 (boundary)	FALSE
M2a	-1648.21	24	3344.41	3490.59		FALSE
M2b	-1642.39	30	3344.78	3527.50	chi2(6) = 11.63, p = 0.071	FALSE
M3	-1647.56	25	3345.12	3497.39		FALSE

M3 uses random intercepts for commune and cercle and fixed effects for region and urban, as set in the configuration after reading these tests.

3.3 The violence effect

Table 4: Violence coefficient (log-odds per unit of $\log(1 + \text{events})$) across models. The design-based check is population-averaged: compare z , not the estimate.

Model	Estimate	SE	z
M2a	-0.249	0.072	-3.46
M2b	-0.226	0.072	-3.14
M3 (final)	-0.248	0.072	-3.45
Design-based (svyglm)	-0.170	0.069	-2.45

In M3, a one-unit increase in $\log(1 + E)$ changes the log-odds of a favorable view by -0.248 (95% CI -0.389 to -0.107, $p < .001$), holding wave, region, urban, and respondent characteristics constant.

Table 5: M3: predicted percent favorable if every respondent's commune had the given number of events, everything else as observed, typical commune and cercle (95% CI).

Events in commune	Predicted % favorable	95% CI
0	83.2	80.1 to 86.2
1	80.7	77.9 to 83.5
5	76.2	72.5 to 80.0
20	70.4	63.4 to 77.3

What the data can and cannot rule out. Going from 0 to 5 events changes predicted favorability by -6.9 percentage points (95% CI -10.9 to -3.0). Effects outside this interval are unlikely; if it includes zero, the data cannot distinguish the association from no association, which is not the same as showing no effect. The smallest 0-to-5-event change this design would detect with 80% power at the 5% level is about 5.6 percentage points (2.8 standard errors of the coefficient, converted with the average slope of the logistic curve).

Design. The log-weight coefficient is 0.080 (SE 0.070); a value more than about two standard errors from zero would suggest informative sampling. The design-based check gives $z = -2.45$ against -3.45 for M3.

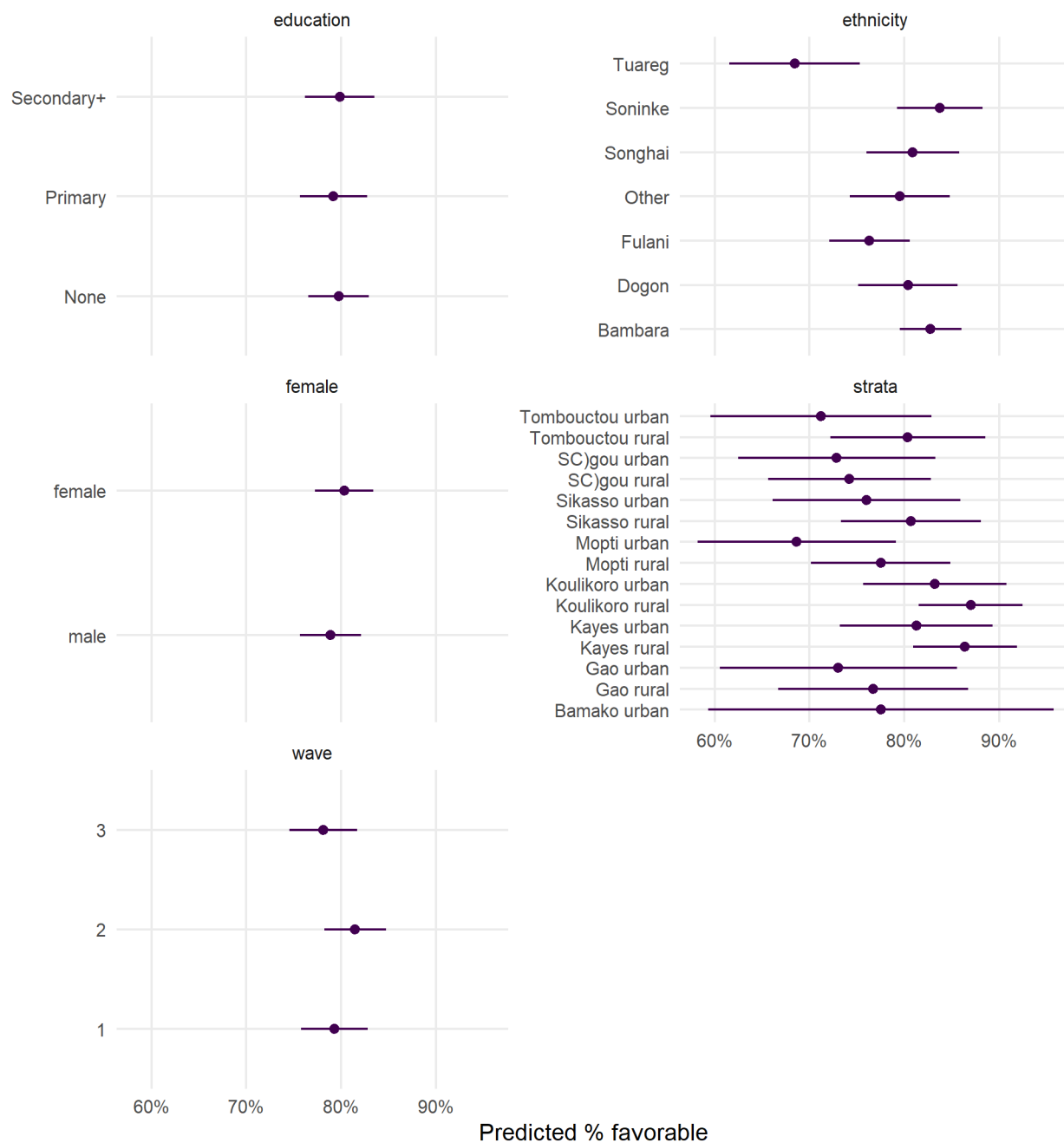


Figure 1: M3: predicted percent favorable (95% CI). Wave and covariates: everyone set to each level, everything else as observed. Strata: average prediction for the people in each stratum.

3.4 Sensitivity: cercle split

Table 6: Violence split into the cercle average and the commune's distance from it (log-odds, SE).

Term	Estimate	SE	z
Cercle average	-0.476	0.133	-3.57
Within cercle	-0.162	0.083	-1.96

The within-cercle coefficient compares communes with more or less violence than other surveyed communes of the same cercle in the same wave; 59.7% of the variation in violence is of this kind. The cercle-average coefficient compares cercles.

4 Diagnostics

Table 7: Collinearity of the M3 predictors (generalized variance inflation factors).

Term	VIF
v	1.05
wave	1.03
region	1.36
urban	1.10
female	1.01
age	1.01
ethnicity	1.15
education	1.01
log_wt_c	1.13

M3 is not singular. Convergence messages from lme4: none.

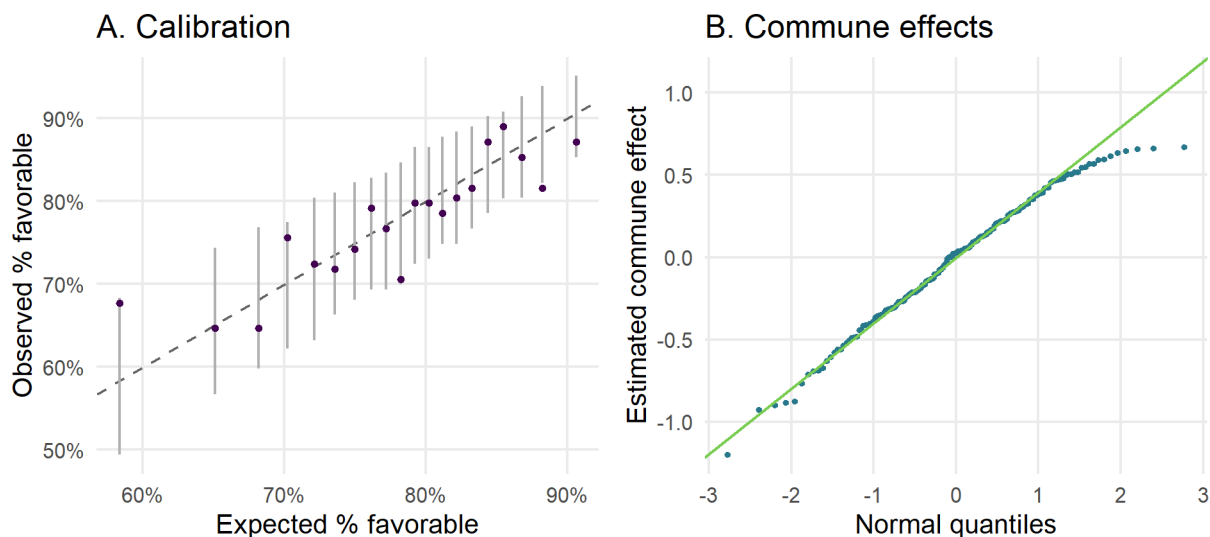


Figure 2: (A) M3 calibration: observed percent favorable in 20 bins of expected probability, against the range the model itself produces (gray, 95% of 200 simulated data sets with new commune and cercle effects). (B) M3 commune random effects against a normal distribution.

5 Maps

Gray communes were not surveyed in that wave: missing data, not zero.

Events of violence against civilians (ACLED) by commune and survey wave

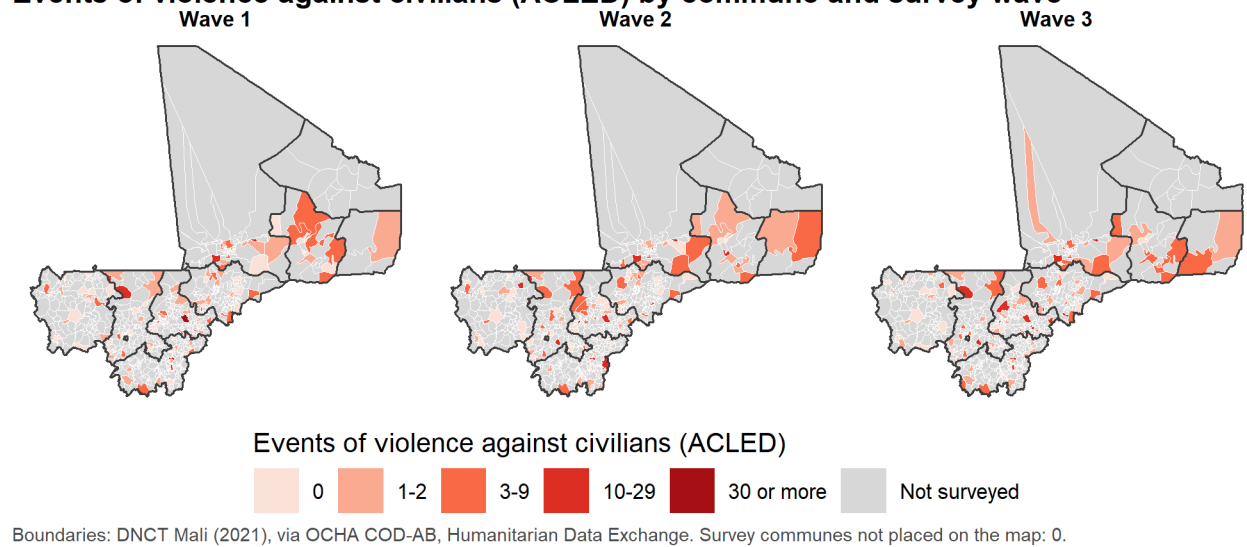


Figure 3: Violence events by commune and wave (surveyed communes only).

Predicted leader favorability by cercle and survey wave

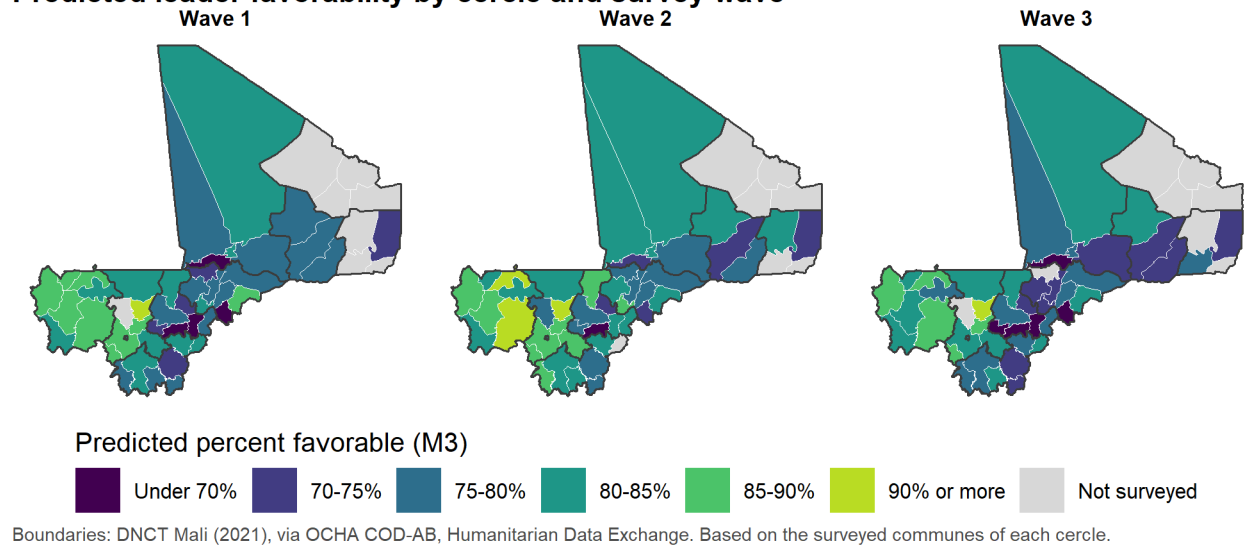


Figure 4: M3 predicted percent favorable by cercle and wave, including each cercle's own random effect, averaged over the respondents of its surveyed communes. It describes the surveyed communes, not the whole cercle.

The cercle map is the easier one to read: each cercle pools several surveyed communes. The commune map below shows more detail, but with eight respondents per commune each value leans heavily on its cercle and on the model, so single communes should not be read on their own.

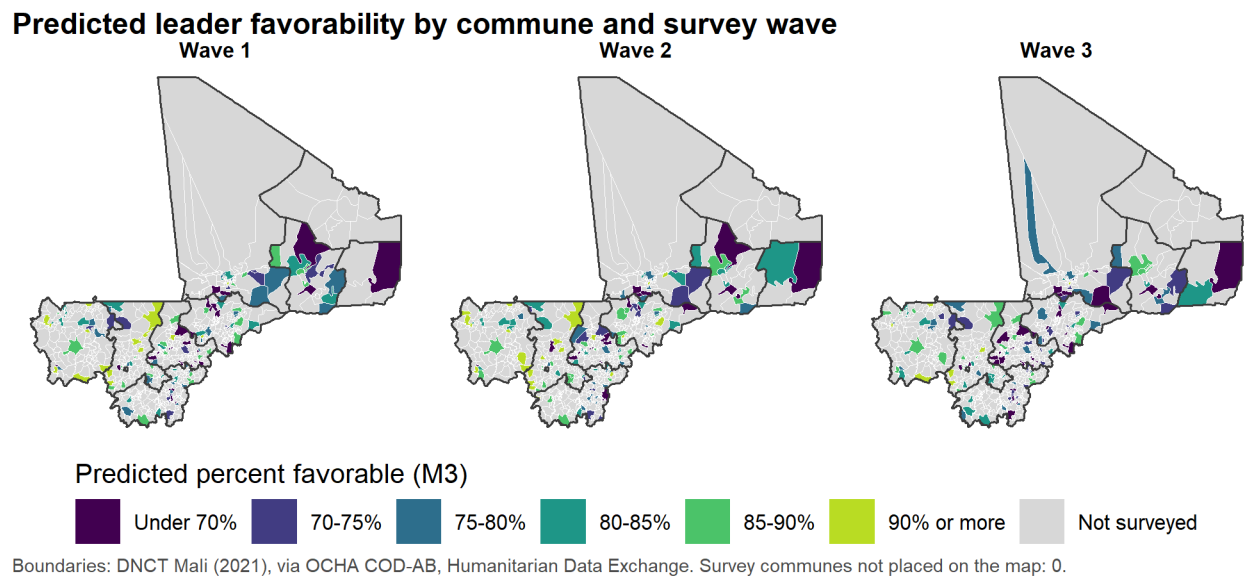


Figure 5: M3 predicted percent favorable by commune and wave, including each commune's and cercle's own random effects.

6 How to explain the result

Among people surveyed in the same region, area, and survey round, predicted favorability toward the leader is about 83.2% where a commune had no violence against civilians and about 76.2% where it had 5 events, a difference of -6.9 percentage points (95% CI -10.9 to -3.0). The data mostly compare different communes, so this is an association, not proof that violence caused the change.

7 Limitations

- **Mostly between communes.** Violence differs mainly between communes. Stable differences the covariates miss could drive part of any association; the cercle-split model removes stable cercle-level context.
- **Unweighted multilevel models.** The log-weight covariate and the design-based check guard against informative sampling but do not replace weighting; commune-level selection weights were not available for a weighted multilevel model.
- **Violence measure.** Only surveyed communes have counts; ACLED undercounts remote areas, which most likely weakens any association. Perpetrator is not available.

- **Sensitive outcome.** Favorability toward the leader may be over-reported where people fear consequences, compressing differences between communes.
- **Coverage.** Insecure areas are harder to survey; estimates describe the population that could be reached.

8 References

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9 Appendix: full coefficient tables

Table 8: Final model M3: all fixed effects (log-odds [95% CI]) and random-effect standard deviations.

	M3
(Intercept)	1.89 [0.74, 3.04]
Violence, log(1 + events)	-0.25 [-0.39, -0.11]
wave: 2	0.15 [-0.08, 0.37]
wave: 3	-0.07 [-0.29, 0.14]
region: Gao	-0.02 [-1.28, 1.24]
region: Kayes	0.12 [-1.08, 1.33]
region: Koulikoro	0.32 [-0.89, 1.52]
region: Mopti	-0.18 [-1.38, 1.01]
region: Sikasso	-0.19 [-1.40, 1.02]
region: SC)gou	-0.44 [-1.64, 0.75]
region: Tombouctou	0.16 [-1.07, 1.38]
urban: urban	-0.34 [-0.68, 0.01]
female: female	0.09 [-0.08, 0.27]
age	0.01 [-0.00, 0.01]
ethnicity: Dogon	-0.16 [-0.51, 0.19]
ethnicity: Fulani	-0.41 [-0.67, -0.15]
ethnicity: Other	-0.22 [-0.55, 0.12]
ethnicity: Songhai	-0.13 [-0.48, 0.22]
ethnicity: Soninke	0.07 [-0.26, 0.40]
ethnicity: Tuareg	-0.82 [-1.17, -0.46]
education: Primary	-0.03 [-0.24, 0.17]
education: Secondary+	0.01 [-0.21, 0.22]
Log weight (centered)	0.08 [-0.06, 0.22]
SD: commune	0.59
SD: cercle	0.41
Num.Obs.	3264
AIC	3345.1
BIC	3497.4

Table 9: M2b: stratum effects (log-odds [95% CI]) against the reference stratum.

	M2b
strata: Gao rural	0.21 [-0.98, 1.40]
strata: Gao urban	1.58 [-0.51, 3.68]
strata: Kayes rural	0.62 [-0.53, 1.77]
strata: Kayes urban	0.10 [-1.20, 1.41]
strata: Koulikoro rural	0.80 [-0.36, 1.96]
strata: Koulikoro urban	0.25 [-1.06, 1.55]
strata: Mopti rural	0.48 [-0.66, 1.61]
strata: Mopti urban	-0.82 [-2.08, 0.44]
strata: Sikasso rural	0.20 [-0.95, 1.34]
strata: Sikasso urban	0.23 [-1.07, 1.53]
strata: SC)gou rural	-0.05 [-1.20, 1.09]
strata: SC)gou urban	-0.19 [-1.49, 1.11]
strata: Tombouctou rural	0.46 [-0.72, 1.65]
strata: Tombouctou urban	0.44 [-0.92, 1.80]